

ClaimLens — Design Document

Author: Capstone Project Candidate | **System:** ClaimLens | **Domain:** Property & Casualty Insurance (P&C;) | **Version:** 1.0.0

1. Problem Statement

Insurance carriers process hundreds of thousands of claims annually across Auto, Home, Renters, Property, and Business coverage lines. Adjusters manually inspect historical files to identify similar cases, assess damage credibility, detect suspicious fraud patterns, and estimate resolution timelines. This manual process causes significant operational latency (average 18–35 days resolution), inconsistent payout decisions, and elevated fraud leakage (estimated at over \$308B annually across the US insurance sector).

2. Solution Overview & Business Objectives

This project provides an intelligent, AI-assisted claims triaging and decisioning microservice. The solution achieves:

- **Automated Hybrid Claim Retrieval:** Finding historical peer cases via in-memory normalized vector embeddings and BM25 keyword matching.
- **Predictive ML Estimation:** Accurately estimating settlement amounts, days to resolution, and fraud probability with explainability.
- **Multi-Agent Orchestration:** Coordinating Investigation, Risk Assessment, and Recommendation agents with LLM-as-Judge validation.
- **Operational Efficiency:** Enabling fast-track approval for low-risk routine claims while routing high-risk claims to SIU.

3. Technology Selection & Justifications

- **OpenAI Models (GPT-5-nano & Text-Embedding-3-Small):** Multi-agent reasoning for underwriting policy synthesis and 1536-dimensional dense claim vector embeddings.
- **In-Memory Normalized Vector Store:** Exact NumPy cosine similarity matrix engine ((N, D) @ (D,)) provides 3.08 ms median matrix search (see docs/latency_benchmark.json) with zero Windows SQLite concurrency lock contention.
- **Rank-BM25 Lexical Engine:** Tokenized BM25Okapi keyword search combined with vector retrieval via Reciprocal Rank Fusion (RRF).
- **Scikit-Learn Ensemble Models:** Random Forest Regressors and Classifiers trained on 10,000 claims sequentially extracted from row group 0 of the Hugging Face 100M dataset provide high-speed, explainable numeric predictions.
- **FastAPI & Streamlit:** FastAPI provides high-throughput asynchronous REST microservices; Streamlit provides a low-latency interactive adjuster UI.
- **LLM-as-Judge & Deterministic Auditing:** Automated evaluation framework quantifying faithfulness, completeness, and underwriting policy compliance.

4. Retrieval Strategy & Hybrid Search

To solve the 'semantic vs. keyword' retrieval dilemma (e.g., matching specific vehicle codes or states vs. abstract loss descriptions), the system implements a hybrid search pipeline:

- **Chunking & Representation:** Each claim is converted into a rich narrative document capturing policy tenure, incident facts, loss amount, deductible, status, and resolution history.
- **Reciprocal Rank Fusion (RRF):** Combines ranked lists with parameter k=60: $RRF_Score = 0.6 / (60 + Rank_Vector) + 0.4 / (60 + Rank_BM25)$.
- **Metadata Filtering:** Enables strict pre- and post-filtering by claim type, state, claim status, payout range, and policyholder claim history.

5. Multi-Agent Workflow Design

The workflow implements an autonomous coordinated reasoning paradigm:

- **Classification & Triage Agent:** Evaluates loss severity, coverage category, and assigns SLA priority tier (P1 Critical to P4 Routine).
- **Investigation Agent:** Gathers policyholder background (tenure, prior claim frequency), verifies incident-to-filing timeline, and clusters similar peer claim precedents.
- **Risk Assessment Agent:** Evaluates financial exposure, calculates claim-to-deductible loss ratios, processes calibrated ML fraud classifier predictions, and extracts top risk drivers.
- **Recommendation Agent:** Determines handling pathways: AUTO_APPROVE (Fast-track), MANUAL_ADJUSTER_REVIEW (Senior review), or SIU_REFERRAL (Special Investigation Unit).
- **LLM-as-Judge Evaluator:** An independent supervisor validating the factual consistency, completeness, and underwriting policy compliance of the generated decision.

6. Evaluation Strategy & Benchmark Results

The solution was evaluated against synthetic benchmark scenarios using the LLM-as-Judge auditing framework and deterministic invariant checks (authoritative source: docs/Evaluation_Summary.json):

Evaluation Metric	Benchmark Score	Evaluation Target	Status
Judge Faithfulness (Fact Grounding)	95.0%	≥ 85.0%	EXCEEDED
Judge Completeness / Relevancy	90.0%	≥ 85.0%	EXCEEDED
Underwriting Policy Compliance	95.0%	≥ 90.0%	EXCEEDED
LLM-as-Judge Quality Score	9.3 / 10.0	≥ 8.0 / 10.0	EXCEEDED
Deterministic Invariants & Guardrails	100.0%	100.0%	PASSED

7. Challenges, Trade-offs & Production Roadmap

- **Challenge:** Operating at 100M claims scale locally.
Decision: Streamed a 10,000-record sequential extract from row group 0 of the Parquet dataset and indexed 2,000 active candidates into the in-memory vector store, achieving 3.08 ms median matrix retrieval and zero disk bloat. Enterprise roadmap targets distributed Qdrant/Milvus clusters.
- **Challenge:** LLM hallucination and policy adherence.
Decision: Implemented input guardrails, strict Pydantic schemas, deterministic financial invariant checks, and an independent LLM-as-Judge audit layer.

- **Future Roadmap:** Vision-LLM integration for vehicle/property damage photograph analysis and automated OCR of police reports.